

Fatemeh Monshizadeh

Email: monshizadehf@cardiff.ac.uk

LinkedIn: <https://www.linkedin.com/in/fatemeh-monshizadeh/>

Research Profile

I am a PhD student at Cardiff Business School, specializing in Digital Twins and healthcare. My research focuses on developing a predictive Digital Twin model to support clinical decision-making in critical care, particularly in optimizing extubation timing for mechanically ventilated patients. The aim is to reduce extubation failure, improve patient outcomes, and enhance critical care unit resource management.

EDUCATION

Cardiff Business School, Cardiff University, UK

2024-Present

PhD, Logistics and Operations Management

Cardiff Business School, Cardiff University, UK

2023-2024

Social Science Research Method (CARBS PhD), Logistics and Operations Management

Thesis: *Application of Digital Twins in Healthcare: Exploring Challenges and Benefits, A Systematic Literature Review.*

Grade: Distinction

University of Tehran, Iran

2016-2019

MSc Industrial Management (Production and Operations Management)

Thesis: Evaluating the Automotive Industry Readiness to implement the Industry4.0 (FCM Approach) (Case Study: Iranian Automotive Industry)

Ferdowsi University of Mashhad, Iran

2011-2015

BSc Public Administration

Thesis: Examining the structure and description of the mission of the Social Insurance Organization

PUBLICATIONS

- **F. Monshizadeh**, MR. Sadeghi Moghadam, T. Mansouri, M. Kumar (2022) *“Developing an Industry 4.0 Readiness Model Using Fuzzy Cognitive Maps (FCM) Approach”*, January 2023, *International Journal of Production Economics*.
- T. Mansouri, MR. Sadeghi Moghadam, **F. Monshizadeh**, A.Zareravasan, *“IoT Data Quality Issues and Potential Solutions: A Literature Review”*, November 2021, *The Computer Journal*.
- **F. Monshizadeh**, Kumar. M, Rostami-Tabar, B, *“Examining Digital Twins Applications, Challenges and Benefits in Healthcare: A Systematic Literature Review”*, IEEE Transactions on Engineering Management, Under review.

TEACHING EXPERIENCES

- **Graduate Tutor**, BS1501- Applied Stats & Maths in Economics & Business, Cardiff Business School, Spring 2026.
- **Step-Up tutor**, Widening Participation, Cardiff University, January 2025-Present.
- **Teaching Assistant**, “Seminar on Quality Management” module, Postgraduate Studies, 10 Students, University of Tehran, Iran. Fall 2017.
- **Teaching Assistant**, “Reengineering Business Process” module, Postgraduate Studies, 10 Students, University of Tehran, Iran. Spring 2016.
- **Teaching Assistant**, “Managing local agencies and municipalities” module, Undergraduate Studies, 22 Students, Ferdowsi University of Mashhad, Iran. Fall 2014.
- **Teaching Assistant**, “Theory of Government Management” module, Undergraduate Studies, 22 Students, Ferdowsi University of Mashhad, Iran. Spring 2015.

WORK EXPERIENCES

Senior Business Development, Investing Company

Jun 2023-Sep 2023

- Analyzing the Application of Digital Twin in Mining Industry

Consultant, ILIA Management Consulting Firm

Mar 2023-Jun 2023

- Writing Private Equity Report
- Collecting primary data and writing CEO Report

Senior Business Analyst, ILIA Management Consulting Firm, doing various projects: Jun 2022-Mar 2023

- Analyzing OpenAI company business model (Rutilea: an AI Company in Japan)
- Entering the Iranian market and providing go to market advice (YnP: Consulting Firm in Russia)
- Analyzing Iran FMCG market (YnP: Consulting Firm in Russia)
- Conducting market research regarding operation monitoring system (Rutilea: an AI Company in Japan)
- Digital transformation research: Analyzing success cases of implementing DT (Rutilea: an AI Company in Japan)

Planning Expert, University of Tehran, Science and Technology Park

July 2020-Jan 2022

- Conducting creative industries research and benchmark
- Doing research and developing productivity model in Iranian companies

HONORS and AWARDS

- **Awarded the NATCOR bursary** to attend the *Optimization Under Uncertainty* course, June 2026, Scotland, UK
- **Best Paper Award**, UKDHC Annual Scientific Informatics Conference, October 2025, UK.
- **Outstanding PhD Peer Mentor Award**, Cardiff Business School, September 2025, Cardiff, UK.
- **Awarded the NATCOR bursary** to attend the *Simulation* course, July 2025, Southampton, UK
- **Best Poster Award**, poster title: *Developing a Conceptual Model to Test Digital Twin in Healthcare*, 4th Welsh Post-Graduate Research Conference (WPGRC) held at Cardiff Business School, June 2024.
- **Cardiff Business School PhD Scholarships Award** in Logistics and Operations Management, July 2023.
- **Ranked 13th** among more than 80,000 participants in the National Graduate Entrance Exam for Industrial Management, Iran, May 2016.

PREFESSIONAL MEMBERSHIP

- **Leadership team member**, DTNetPlus: SIG LLODIA - Advancing Digital Twin Interoperability
- **Student Member**, The Operational Research Society (The OR Society)

CONFERENCES

- **EurOMA 2026**, June 27th - July 2nd, Vienna, Austria
- **Perspectives on the Societal Impact of Digital Twins Conference**, DTNet+ research community, Cambridge, UK, March 2026.
- **2nd UK Digital Health and Care**, Annual Scientific Informatics Conference, October 2025.
- **Gwalior**, an Institute of National Importance under the Ministry of Education, Government of India, International conference on AI-driven human-centric solutions and sustainable transformation (AI-HCSST 2025), March 2025.
- **Cardiff Business School**, 4th Welsh Post-Graduate Research Conference (WPGRC), June 2024.

SKILLS

Communication; I have good written and verbal communication skills, refined through my research experience and academic training. I am adept at adjusting my communication style for various audiences, ensuring that complex information is presented clearly and understandably.

Computer Skills; SPSS, Smart PLS, R, and Microsoft Office.